

Appendix D

Thermodynamic Properties of H₂O

Table D.1 Saturation Properties of Water and Steam: Temperature Increments

<i>T</i> (K)	<i>P</i> (kPa)	<i>v</i> (m ³ /kg) sat. liquid	<i>v</i> (m ³ /kg) sat. vapor	<i>u</i> (kJ/kg) sat. liquid	<i>u</i> (kJ/kg) sat. vapor	<i>h</i> (kJ/kg) sat. liquid	<i>h</i> (kJ/kg) sat. vapor	<i>s</i> (kJ/kg·K) sat. liquid	<i>s</i> (kJ/kg·K) sat. vapor
273.16	0.611650	0.0010002	205.99	0	2374.9	0.00061	2500.9	0	9.1555
274	0.650030	0.0010002	194.43	3.5435	2376.1	3.5442	2502.5	0.012952	9.1331
275	0.698460	0.0010001	181.60	7.7590	2377.5	7.7597	2504.3	0.028309	9.1066
276	0.750070	0.0010001	169.71	11.971	2378.8	11.972	2506.1	0.043600	9.0804
277	0.805020	0.0010001	158.70	16.181	2380.2	16.182	2508.0	0.058825	9.0544
278	0.863500	0.0010001	148.48	20.388	2381.6	20.389	2509.8	0.073985	9.0287
279	0.925700	0.0010001	139.00	24.593	2383.0	24.594	2511.6	0.089083	9.0031
280	0.991830	0.0010001	130.19	28.795	2384.3	28.796	2513.4	0.10412	8.9779
281	1.062200	0.0010002	122.01	32.996	2385.7	32.997	2515.3	0.11909	8.9528
282	1.136800	0.0010003	114.40	37.194	2387.1	37.195	2517.1	0.13401	8.9280
283	1.216000	0.0010003	107.32	41.391	2388.4	41.392	2518.9	0.14886	8.9034
284	1.300000	0.0010004	100.74	45.586	2389.8	45.587	2520.8	0.16366	8.8791
285	1.389100	0.0010005	94.602	49.779	2391.2	49.780	2522.6	0.17840	8.8549
286	1.483600	0.0010006	88.887	53.971	2392.6	53.973	2524.4	0.19308	8.8310
287	1.583600	0.0010008	83.560	58.162	2393.9	58.163	2526.2	0.20771	8.8073
288	1.689500	0.0010009	78.592	62.351	2395.3	62.353	2528.1	0.22228	8.7838
289	1.801600	0.0010011	73.955	66.540	2396.7	66.542	2529.9	0.23680	8.7605
290	1.920100	0.0010012	69.625	70.727	2398.0	70.729	2531.7	0.25126	8.7374
291	2.045400	0.0010014	65.581	74.914	2399.4	74.916	2533.5	0.26568	8.7145
292	2.177900	0.0010016	61.801	79.099	2400.8	79.101	2535.3	0.28003	8.6918
293	2.317800	0.0010018	58.267	83.284	2402.1	83.286	2537.2	0.29434	8.6693
294	2.465500	0.0010020	54.960	87.468	2403.5	87.471	2539.0	0.30860	8.6471
295	2.621300	0.0010022	51.865	91.652	2404.8	91.654	2540.8	0.32280	8.6250
296	2.785700	0.0010025	48.966	95.835	2406.2	95.837	2542.6	0.33696	8.6031
297	2.959100	0.0010027	46.251	100.02	2407.6	100.02	2544.4	0.35106	8.5814
298	3.141800	0.0010030	43.705	104.20	2408.9	104.20	2546.2	0.36512	8.5599
299	3.334300	0.0010032	41.318	108.38	2410.3	108.38	2548.0	0.37913	8.5385
300	3.536900	0.0010035	39.078	112.56	2411.6	112.56	2549.9	0.39309	8.5174
301	3.750200	0.0010038	36.976	116.74	2413.0	116.75	2551.7	0.40700	8.4964
302	3.974600	0.0010041	35.002	120.92	2414.4	120.93	2553.5	0.42087	8.4756
303	4.210600	0.0010044	33.147	125.10	2415.7	125.11	2555.3	0.43469	8.4550
304	4.458700	0.0010047	31.403	129.28	2417.1	129.29	2557.1	0.44846	8.4346
305	4.719400	0.0010050	29.764	133.46	2418.4	133.47	2558.9	0.46219	8.4144
306	4.993299	0.0010053	28.222	137.64	2419.8	137.65	2560.7	0.47587	8.3943
307	5.280799	0.0010057	26.770	141.82	2421.1	141.83	2562.5	0.48950	8.3744
308	5.582599	0.0010060	25.403	146.00	2422.5	146.01	2564.3	0.50310	8.3546
309	5.899199	0.0010063	24.116	150.18	2423.8	150.19	2566.1	0.51664	8.3351
310	6.231199	0.0010067	22.903	154.36	2425.2	154.37	2567.9	0.53015	8.3156
311	6.579299	0.0010071	21.759	158.54	2426.5	158.55	2569.7	0.54361	8.2964
312	6.944099	0.0010074	20.680	162.72	2427.8	162.73	2571.5	0.55702	8.2773
313	7.326199	0.0010078	19.663	166.90	2429.2	166.91	2573.2	0.57040	8.2584
314	7.726299	0.0010082	18.702	171.08	2430.5	171.09	2575.0	0.58373	8.2396
315	8.145199	0.0010086	17.795	175.26	2431.9	175.27	2576.8	0.59702	8.2210
316	8.583499	0.0010090	16.938	179.44	2433.2	179.45	2578.6	0.61027	8.2025
317	9.041899	0.0010094	16.129	183.62	2434.5	183.63	2580.4	0.62348	8.1842
318	9.521299	0.0010099	15.363	187.80	2435.9	187.81	2582.2	0.63664	8.1660
319	10.022989	0.0010103	14.639	191.98	2437.2	191.99	2583.9	0.64977	8.1480
320	10.546989	0.0010107	13.954	196.16	2438.5	196.17	2585.7	0.66285	8.1302

<i>T</i> (K)	<i>P</i> (MPa)	<i>v</i> (m ³ /kg) sat. liquid	<i>v</i> (m ³ /kg) sat. vapor	<i>u</i> (kJ/kg) sat. liquid	<i>u</i> (kJ/kg) sat. vapor	<i>h</i> (kJ/kg) sat. liquid	<i>h</i> (kJ/kg) sat. vapor	<i>s</i> (kJ/kg·K) sat. liquid	<i>s</i> (kJ/kg·K) sat. vapor
320	0.010547	0.0010107	13.954	196.16	2438.5	196.17	2585.7	0.66285	8.1302
325	0.013532	0.0010130	11.039	217.07	2445.2	217.08	2594.6	0.72768	8.0430
330	0.017214	0.0010155	8.8050	237.98	2451.8	238.00	2603.3	0.79154	7.9592
335	0.021719	0.0010181	7.0788	258.9	2458.3	258.93	2612.1	0.85447	7.8787
340	0.027189	0.0010209	5.7339	279.84	2464.8	279.87	2620.7	0.91650	7.8013
345	0.033784	0.0010239	4.6776	300.79	2471.2	300.82	2629.3	0.97766	7.7267
350	0.041683	0.0010270	3.8419	321.75	2477.6	321.79	2637.7	1.0380	7.6549
355	0.051081	0.0010303	3.1759	342.73	2483.9	342.78	2646.1	1.0975	7.5857
360	0.062195	0.0010337	2.6414	363.73	2490.1	363.79	2654.4	1.1562	7.5190
365	0.075261	0.0010373	2.2098	384.75	2496.2	384.82	2662.5	1.2142	7.4545
370	0.090536	0.0010410	1.8590	405.79	2502.3	405.88	2670.6	1.2715	7.3923
375	0.108310	0.0010449	1.5722	426.86	2508.2	426.97	2678.5	1.3281	7.3321
380	0.128860	0.0010490	1.3364	447.96	2514.1	448.09	2686.2	1.3839	7.2738
385	0.152529	0.0010532	1.1414	469.09	2519.8	469.25	2693.9	1.4392	7.2174
390	0.179649	0.0010575	0.97928	490.25	2525.4	490.44	2701.3	1.4938	7.1627
395	0.210609	0.0010620	0.84389	511.45	2530.9	511.67	2708.6	1.5478	7.1097
400	0.245779	0.0010667	0.73024	532.69	2536.2	532.95	2715.7	1.6013	7.0581
405	0.285589	0.0010715	0.63441	553.98	2541.4	554.28	2722.6	1.6541	7.0081
410	0.330459	0.0010765	0.55323	575.31	2546.5	575.66	2729.3	1.7065	6.9593
415	0.380879	0.0010817	0.48418	596.69	2551.4	597.10	2735.8	1.7583	6.9119
420	0.437309	0.0010870	0.42520	618.13	2556.2	618.60	2742.1	1.8097	6.8656
425	0.500259	0.0010926	0.37463	639.62	2560.7	640.17	2748.1	1.8606	6.8205
430	0.570269	0.0010983	0.33110	661.18	2565.1	661.80	2753.9	1.9110	6.7764
435	0.647879	0.0011042	0.29350	682.8	2569.3	683.52	2759.5	1.9610	6.7333
440	0.733679	0.0011103	0.26090	704.5	2573.3	705.31	2764.7	2.0106	6.6911
445	0.828249	0.0011166	0.23255	726.26	2577.1	727.19	2769.7	2.0598	6.6498
450	0.932209	0.0011232	0.20781	748.11	2580.7	749.16	2774.4	2.1087	6.6092
455	1.046289	0.0011299	0.18616	770.05	2584.0	771.23	2778.8	2.1571	6.5694
460	1.170988	0.0011369	0.16715	792.07	2587.2	793.41	2782.9	2.2053	6.5303
465	1.306987	0.0011442	0.15041	814.2	2590.1	815.69	2786.6	2.2532	6.4917
470	1.455187	0.0011517	0.13564	836.42	2592.7	838.09	2790.0	2.3007	6.4538
475	1.616086	0.0011594	0.12255	858.75	2595.0	860.62	2793.1	2.3480	6.4164
480	1.790586	0.0011675	0.11094	881.19	2597.1	883.28	2795.8	2.3950	6.3794
485	1.979285	0.0011758	0.10061	903.76	2598.9	906.09	2798.1	2.4418	6.3428
490	2.183184	0.0011845	0.091390	926.45	2600.5	929.04	2800.0	2.4884	6.3066
495	2.402884	0.0011935	0.083149	949.28	2601.7	952.15	2801.4	2.5348	6.2708
500	2.639283	0.0012029	0.075764	972.26	2602.5	975.43	2802.5	2.5810	6.2351
505	2.893182	0.0012127	0.069131	995.38	2603.0	998.89	2803.1	2.6271	6.1997
510	3.165582	0.0012228	0.063161	1018.7	2603.2	1022.5	2803.2	2.6731	6.1645
515	3.457181	0.0012334	0.057776	1042.1	2603.0	1046.4	2802.7	2.7189	6.1293
520	3.769080	0.0012445	0.052910	1065.8	2602.4	1070.5	2801.8	2.7647	6.0942
525	4.101980	0.0012561	0.048503	1089.6	2601.4	1094.8	2800.3	2.8104	6.0591
530	4.456979	0.0012682	0.044503	1113.7	2599.9	1119.3	2798.2	2.8561	6.0239
535	4.834978	0.0012809	0.040868	1138	2597.9	1144.2	2795.5	2.9019	5.9885
540	5.236977	0.0012942	0.037556	1162.5	2595.5	1169.3	2792.2	2.9476	5.9530
545	5.664076	0.0013083	0.034535	1187.3	2592.5	1194.7	2788.1	2.9935	5.9171
550	6.117276	0.0013231	0.031772	1212.4	2588.9	1220.5	2783.3	3.0394	5.8809
555	6.597675	0.0013387	0.029242	1237.8	2584.8	1246.6	2777.7	3.0855	5.8443
560	7.106274	0.0013553	0.026920	1263.5	2579.9	1273.1	2771.2	3.1319	5.8071
565	7.644473	0.0013729	0.024786	1289.6	2574.4	1300.1	2763.9	3.1785	5.7693
570	8.213272	0.0013917	0.022820	1316	2568.0	1327.5	2755.5	3.2254	5.7307
575	8.814071	0.0014118	0.021005	1343	2560.9	1355.4	2746.0	3.2727	5.6912
580	9.448070	0.0014334	0.019328	1370.4	2552.7	1383.9	2735.3	3.3205	5.6506
585	10.117686	0.0014567	0.017773	1398.4	2543.5	1413.1	2723.3	3.3690	5.6087
590	10.821674	0.0014820	0.016329	1426.9	2533.2	1443.0	2709.9	3.4181	5.5654
595	11.563662	0.0015095	0.014984	1456.2	2521.5	1473.7	2694.8	3.4680	5.5203
600	12.345649	0.0015399	0.013728	1486.4	2508.3	1505.4	2677.8	3.5190	5.4731
605	13.167635	0.0015735	0.012552	1517.4	2493.4	1538.1	2658.7	3.5713	5.4234
610	14.033620	0.0016112	0.011446	1549.6	2476.4	1572.2	2637.0	3.6252	5.3707
615	14.943604	0.0016541	0.010400	1583.2	2456.9	1608.0	2612.3	3.6811	5.3142
620	15.901586	0.0017039	0.0094067	1618.6	2434.3	1645.7	2583.9	3.7396	5.2528
625	16.908567	0.0017634	0.0084538	1656.5	2407.8	1686.3	2550.7	3.8019	5.1851
630	17.969544	0.0018374	0.0075279	1697.7	2375.8	1730.7	2511.1	3.8698	5.1084
635	19.086516	0.0019353	0.0066074	1744.3	2335.7	1781.2	2461.8	3.9463	5.0181
640	20.265481	0.0020767	0.0056451	1799.7	2281.1	1841.8	2395.5	4.0375	4.9027
645	21.515425	0.0023527	0.0044553	1880.5	2185.2	1931.1	2281.0	4.1722	4.7147
647.096	22.064000	0.0031056	0.0031056	2015.7	2015.7	2084.3	2084.3	4.4070	4.4070

Table D.2 Saturation Properties of Water and Steam: Pressure Increments

P (kPa)	T (K)	v (m ³ /kg) sat. liquid	v (m ³ /kg) sat. vapor	u (kJ/kg) sat. liquid	u (kJ/kg) sat. vapor	h (kJ/kg) sat. liquid	h (kJ/kg) sat. vapor	s (kJ/kg·K) sat. liquid	s (kJ/kg·K) sat. vapor
2.0	290.64	0.0010014	66.987	73.426	2398.9	73.428	2532.9	0.26056	8.7226
4.0	302.11	0.0010041	34.791	121.38	2414.5	121.39	2553.7	0.42239	8.4734
6.0	309.31	0.0010065	23.733	151.47	2424.2	151.48	2566.6	0.52082	8.3290
8.0	314.66	0.0010085	18.099	173.83	2431.4	173.84	2576.2	0.59249	8.2273
10.0	318.96	0.0010103	14.670	191.80	2437.2	191.81	2583.9	0.6492	8.1488
12.0	322.57	0.0010119	12.358	206.90	2442.0	206.91	2590.3	0.69628	8.0849
14.0	325.70	0.0010134	10.691	219.98	2446.1	219.99	2595.8	0.73664	8.0311
16.0	328.46	0.0010147	9.4306	231.55	2449.8	231.57	2600.6	0.77201	7.9846
18.0	330.95	0.0010160	8.4431	241.95	2453.0	241.96	2605.0	0.80355	7.9437
20.0	333.21	0.0010172	7.6480	251.40	2456.0	251.42	2608.9	0.83202	7.9072
22.0	335.28	0.0010183	6.9936	260.09	2458.7	260.11	2612.5	0.85800	7.8743
24.0	337.20	0.0010193	6.4453	268.13	2461.2	268.15	2615.9	0.88191	7.8442
26.0	338.99	0.0010203	5.9792	275.62	2463.5	275.64	2619.0	0.90407	7.8167
28.0	340.67	0.0010213	5.5778	282.64	2465.7	282.66	2621.8	0.92472	7.7912
30.0	342.25	0.0010222	5.2284	289.24	2467.7	289.27	2624.5	0.94407	7.7675
32.0	343.74	0.0010231	4.9215	295.49	2469.6	295.52	2627.1	0.96228	7.7453
34.0	345.15	0.0010240	4.6497	301.41	2471.4	301.45	2629.5	0.97948	7.7246
36.0	346.50	0.0010248	4.4072	307.05	2473.1	307.09	2631.8	0.99579	7.7050
38.0	347.78	0.0010256	4.1895	312.43	2474.8	312.47	2634.0	1.0113	7.6865
40.0	349.01	0.0010264	3.9930	317.58	2476.3	317.62	2636.1	1.0261	7.6690
42.0	350.18	0.0010271	3.8146	322.52	2477.8	322.56	2638.0	1.0402	7.6524
44.0	351.32	0.0010279	3.6520	327.27	2479.3	327.31	2639.9	1.0537	7.6365
46.0	352.40	0.0010286	3.5031	331.83	2480.6	331.88	2641.8	1.0667	7.6214
48.0	353.45	0.0010293	3.3663	336.24	2481.9	336.29	2643.5	1.0792	7.6069
50.0	354.47	0.0010299	3.2400	340.49	2483.2	340.54	2645.2	1.0912	7.5930
52.0	355.45	0.0010306	3.1232	344.60	2484.4	344.66	2646.8	1.1028	7.5797
54.0	356.40	0.0010312	3.0148	348.59	2485.6	348.64	2648.4	1.1140	7.5669
56.0	357.32	0.0010319	2.9139	352.45	2486.8	352.51	2649.9	1.1248	7.5545
58.0	358.21	0.0010325	2.8198	356.20	2487.9	356.26	2651.4	1.1353	7.5426
60.0	359.08	0.0010331	2.7317	359.84	2489.0	359.91	2652.9	1.1454	7.5311
62.0	359.92	0.0010337	2.6492	363.39	2490.0	363.45	2654.2	1.1553	7.5200
64.0	360.74	0.0010342	2.5716	366.84	2491.0	366.91	2655.6	1.1649	7.5093
66.0	361.54	0.0010348	2.4986	370.20	2492.0	370.27	2656.9	1.1742	7.4989
68.0	362.32	0.0010354	2.4298	373.48	2493.0	373.55	2658.2	1.1833	7.4888
70.0	363.08	0.0010359	2.3648	376.68	2493.9	376.75	2659.4	1.1921	7.4790
72.0	363.82	0.0010364	2.3033	379.80	2494.8	379.88	2660.6	1.2007	7.4695
74.0	364.55	0.0010370	2.2450	382.86	2495.7	382.94	2661.8	1.2091	7.4602
76.0	365.26	0.0010375	2.1897	385.84	2496.5	385.92	2663.0	1.2172	7.4512
78.0	365.96	0.0010380	2.1371	388.77	2497.4	388.85	2664.1	1.2252	7.4425
80.0	366.64	0.0010385	2.0871	391.63	2498.2	391.71	2665.2	1.2330	7.4339
82.0	367.30	0.0010390	2.0394	394.43	2499.0	394.51	2666.3	1.2407	7.4256
84.0	367.95	0.0010395	1.9940	397.18	2499.8	397.26	2667.3	1.2482	7.4175
86.0	368.59	0.001040	1.9506	399.87	2500.6	399.96	2668.3	1.2555	7.4096
88.0	369.22	0.0010404	1.9091	402.51	2501.3	402.60	2669.3	1.2626	7.4018
90.0	369.84	0.0010409	1.8694	405.10	2502.1	405.20	2670.3	1.2696	7.3943
92.0	370.44	0.0010414	1.8313	407.65	2502.8	407.75	2671.3	1.2765	7.3869
94.0	371.04	0.0010418	1.7949	410.15	2503.5	410.25	2672.2	1.2833	7.3796
96.0	371.62	0.0010423	1.7599	412.61	2504.2	412.71	2673.1	1.2899	7.3726
98.0	372.19	0.0010427	1.7262	415.02	2504.9	415.13	2674.1	1.2964	7.3656
100.0	372.76	0.0010432	1.6939	417.40	2505.6	417.50	2674.9	1.3028	7.3588

<i>P</i> (MPa)	<i>T</i> (K)	<i>v</i> (m ³ /kg) sat. liquid	<i>v</i> (m ³ /kg) sat. vapor	<i>u</i> (kJ/kg) sat. liquid	<i>u</i> (kJ/kg) sat. vapor	<i>h</i> (kJ/kg) sat. liquid	<i>h</i> (kJ/kg) sat. vapor	<i>s</i> (kJ/kg · K) sat. liquid	<i>s</i> (kJ/kg · K) sat. vapor
0.10	372.76	0.0010432	1.6939	417.40	2505.6	417.50	2674.9	1.3028	7.3588
0.20	393.36	0.0010605	0.88568	504.49	2529.1	504.70	2706.2	1.5302	7.1269
0.30	406.67	0.0010732	0.60576	561.10	2543.2	561.43	2724.9	1.6717	6.9916
0.40	416.76	0.0010836	0.46238	604.22	2553.1	604.65	2738.1	1.7765	6.8955
0.50	424.98	0.0010925	0.37481	639.54	2560.7	640.09	2748.1	1.8604	6.8207
0.60	431.98	0.0011006	0.31558	669.72	2566.8	670.38	2756.1	1.9308	6.7592
0.70	438.10	0.0011080	0.27277	696.23	2571.8	697.00	2762.8	1.9918	6.7071
0.80	443.56	0.0011148	0.24034	719.97	2576.0	720.86	2768.3	2.0457	6.6616
0.90	448.50	0.0011212	0.21489	741.55	2579.6	742.56	2773.0	2.0940	6.6213
1.00	453.03	0.0011272	0.19436	761.39	2582.7	762.52	2777.1	2.1381	6.5850
1.10	457.21	0.0011330	0.17745	779.78	2585.5	781.03	2780.6	2.1785	6.5520
1.20	461.11	0.0011385	0.16326	796.96	2587.8	798.33	2783.7	2.2159	6.5217
1.30	464.75	0.0011438	0.15119	813.11	2589.9	814.60	2786.5	2.2508	6.4936
1.40	468.19	0.0011489	0.14078	828.36	2591.8	829.97	2788.8	2.2835	6.4675
1.50	471.44	0.0011539	0.13171	842.83	2593.4	844.56	2791.0	2.3143	6.4430
1.60	474.52	0.0011587	0.12374	856.60	2594.8	858.46	2792.8	2.3435	6.4199
1.70	477.46	0.0011634	0.11667	869.76	2596.1	871.74	2794.5	2.3711	6.3981
1.80	480.26	0.0011679	0.11037	882.37	2597.2	884.47	2795.9	2.3975	6.3775
1.90	482.95	0.0011724	0.10470	894.48	2598.2	896.71	2797.2	2.4227	6.3578
2.00	485.53	0.0011767	0.099585	906.14	2599.1	908.50	2798.3	2.4468	6.3390
2.10	488.01	0.0011810	0.094938	917.39	2599.9	919.87	2799.3	2.4699	6.3210
2.20	490.4	0.0011852	0.090698	928.27	2600.6	930.87	2800.1	2.4921	6.3038
2.30	492.71	0.0011894	0.086815	938.79	2601.1	941.53	2800.8	2.5136	6.2872
2.40	494.94	0.0011934	0.083244	949.00	2601.6	951.87	2801.4	2.5343	6.2712
2.50	497.10	0.0011974	0.079949	958.91	2602.1	961.91	2801.9	2.5543	6.2558
2.60	499.20	0.0012014	0.076899	968.55	2602.4	971.67	2802.3	2.5736	6.2409
2.70	501.23	0.0012053	0.074066	977.93	2602.7	981.18	2802.7	2.5924	6.2264
2.80	503.21	0.0012091	0.071429	987.07	2602.9	990.46	2802.9	2.6106	6.2124
2.90	505.13	0.0012129	0.068968	995.99	2603.1	999.51	2803.1	2.6283	6.1988
3.00	507.00	0.0012167	0.066664	1004.7	2603.2	1008.3	2803.2	2.6455	6.1856
3.10	508.83	0.0012204	0.064504	1013.2	2603.2	1017.0	2803.2	2.6623	6.1727
3.20	510.61	0.0012241	0.062475	1021.5	2603.2	1025.4	2803.1	2.6787	6.1602
3.30	512.35	0.0012278	0.060564	1029.7	2603.2	1033.7	2803.0	2.6946	6.1479
3.40	514.05	0.0012314	0.058761	1037.7	2603.1	1041.8	2802.9	2.7102	6.1360
3.50	515.71	0.0012350	0.057058	1045.5	2602.9	1049.8	2802.6	2.7254	6.1243
3.60	517.33	0.0012385	0.055446	1053.1	2602.8	1057.6	2802.4	2.7403	6.1129
3.70	518.92	0.0012421	0.053918	1060.7	2602.6	1065.3	2802.1	2.7549	6.1018
3.80	520.48	0.0012456	0.052467	1068.1	2602.3	1072.8	2801.7	2.7691	6.0908
3.90	522.01	0.0012491	0.051089	1075.3	2602.0	1080.2	2801.3	2.7831	6.0801
4.00	523.50	0.0012526	0.049776	1082.5	2601.7	1087.5	2800.8	2.7968	6.0696
4.10	524.97	0.0012560	0.048525	1089.5	2601.4	1094.7	2800.3	2.8102	6.0592
4.20	526.41	0.0012594	0.047332	1096.4	2601.0	1101.7	2799.8	2.8234	6.0491
4.30	527.83	0.0012629	0.046192	1103.2	2600.6	1108.7	2799.2	2.8363	6.0391
4.40	529.22	0.0012663	0.045102	1109.9	2600.1	1115.5	2798.6	2.8490	6.0293
4.50	530.59	0.0012696	0.044059	1116.5	2599.7	1122.2	2797.9	2.8615	6.0197
4.60	531.93	0.0012730	0.043059	1123.0	2599.2	1128.9	2797.3	2.8738	6.0102
4.70	533.25	0.0012764	0.042100	1129.5	2598.7	1135.5	2796.5	2.8859	6.0009
4.80	534.55	0.0012797	0.041180	1135.8	2598.1	1141.9	2795.8	2.8978	5.9917
4.90	535.83	0.0012831	0.040296	1142.0	2597.6	1148.3	2795.0	2.9095	5.9826
5.00	537.09	0.0012864	0.039446	1148.2	2597.0	1154.6	2794.2	2.9210	5.9737

(continued)

Table D.2 (continued)

P (MPa)	T (K)	v (m ³ /kg) sat. liquid	v (m ³ /kg) sat. vapor	u (kJ/kg) sat. liquid	u (kJ/kg) sat. vapor	h (kJ/kg) sat. liquid	h (kJ/kg) sat. vapor	s (kJ/kg·K) sat. liquid	s (kJ/kg·K) sat. vapor
5.0	537.09	0.0012864	0.039446	1148.2	2597.0	1154.6	2794.2	2.9210	5.9737
5.5	543.12	0.0013029	0.035642	1177.9	2593.7	1185.1	2789.7	2.9762	5.9307
6.0	548.73	0.0013193	0.032448	1206.0	2589.9	1213.9	2784.6	3.0278	5.8901
6.5	554.01	0.0013356	0.029727	1232.7	2585.7	1241.4	2778.9	3.0764	5.8516
7.0	558.98	0.0013519	0.027378	1258.2	2581	1267.7	2772.6	3.1224	5.8148
7.5	563.69	0.0013682	0.025330	1282.7	2575.9	1292.9	2765.9	3.1662	5.7793
8.0	568.16	0.0013847	0.023526	1306.2	2570.5	1317.3	2758.7	3.2081	5.7450
8.5	572.42	0.0014013	0.021923	1329.0	2564.7	1340.9	2751.0	3.2483	5.7117
9.0	576.49	0.0014181	0.020490	1351.1	2558.5	1363.9	2742.9	3.2870	5.6791
9.5	580.40	0.0014352	0.019199	1372.6	2552.0	1386.2	2734.4	3.3244	5.6473
10.0	584.15	0.0014526	0.018030	1393.5	2545.2	1408.1	2725.5	3.3606	5.6160
10.5	587.75	0.0014703	0.016965	1414.0	2538.0	1429.4	2716.1	3.3959	5.5851
11.0	591.23	0.0014885	0.015990	1434.1	2530.5	1450.4	2706.3	3.4303	5.5545
11.5	594.58	0.0015071	0.015093	1453.8	2522.6	1471.1	2696.1	3.4638	5.5241
12.0	597.83	0.0015263	0.014264	1473.1	2514.3	1491.5	2685.4	3.4967	5.4939
12.5	600.96	0.0015461	0.013496	1492.3	2505.6	1511.6	2674.3	3.5290	5.4638
13.0	604.00	0.0015665	0.012780	1511.1	2496.5	1531.5	2662.7	3.5608	5.4336
13.5	606.95	0.0015877	0.012112	1529.9	2487.0	1551.3	2650.5	3.5921	5.4032
14.0	609.82	0.0016097	0.011485	1548.4	2477.1	1571.0	2637.9	3.6232	5.3727
14.5	612.60	0.0016328	0.010895	1566.9	2466.6	1590.6	2624.6	3.6539	5.3418
15.0	615.31	0.0016570	0.010338	1585.3	2455.6	1610.2	2610.7	3.6846	5.3106
15.5	617.94	0.0016824	0.0098106	1603.8	2444.1	1629.9	2596.1	3.7151	5.2788
16.0	620.50	0.0017094	0.0093088	1622.3	2431.8	1649.7	2580.8	3.7457	5.2463
16.5	623.00	0.0017383	0.0088299	1641.0	2418.9	1669.7	2564.6	3.7765	5.2130
17.0	625.44	0.0017693	0.0083709	1659.9	2405.2	1690.0	2547.5	3.8077	5.1787
17.5	627.82	0.0018029	0.0079292	1679.2	2390.5	1710.8	2529.3	3.8394	5.1431
18.0	630.14	0.0018398	0.0075017	1699.0	2374.8	1732.1	2509.8	3.8718	5.1061
18.5	632.41	0.0018807	0.0070856	1719.3	2357.8	1754.1	2488.8	3.9053	5.0670
19.0	634.62	0.0019268	0.0066773	1740.5	2339.1	1777.2	2466.0	3.9401	5.0256
19.5	636.79	0.0019792	0.0062725	1762.8	2318.5	1801.4	2440.8	3.9767	4.9808
20.0	638.90	0.002040	0.0058652	1786.4	2295.0	1827.2	2412.3	4.0156	4.9314
20.5	640.96	0.0021126	0.0054457	1812.0	2267.6	1855.3	2379.2	4.0579	4.8753
21.0	642.98	0.0022055	0.0049961	1841.2	2233.7	1887.6	2338.6	4.1064	4.8079
21.5	644.94	0.0023468	0.0044734	1879.1	2186.9	1929.5	2283.1	4.1698	4.7181
22.0	646.86	0.0027044	0.0036475	1951.8	2092.8	2011.3	2173.1	4.2945	4.5446
22.064	647.096	0.0031056	0.0031056	2015.7	2015.7	2084.3	2084.3	4.4070	4.4070